

Notes: Bhattacharya, Hackethal, Kaesler, Loos, and Meyer (RFS, 2012)

Is Unbiased Financial Advice to Retail Investors Sufficient? Answers from a Large Field Study

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Field setting and timeline	4
3.2	Advice algorithm and recommended portfolios	4
3.3	Degree of following	4
3.4	Portfolio-efficiency outcomes	4
3.5	Investor characteristics	5
4	Models and empirical specifications	5
4.1	Probit model for obtaining advice	5
4.2	OLS model for following advice	5
4.3	Advice-quality regression	6
4.4	Average-advisee regression	6
4.5	Partial-following regression	6
5	Identification and empirical design	6
5.1	What makes the field study useful	6
5.2	Main identification limits	7
5.3	Policy interpretation	7
6	Tables and figures	7
6.1	Table 1: Advisees' original portfolios versus recommended portfolios	7
6.2	Table 2 and Figure 1: recommended asset classes by risk aversion, and field-study timeline . .	8
6.3	Table 3 and Table 4: data collected and measurement of the degree of following	9
6.4	Table 5: Summary statistics	9
6.5	Figure 2 and Table 6: Initial degree of following and post-advice implementation	10
6.6	Table 7: Who opts for advice? A probit test	11
6.7	Table 8: Who chooses to follow advice?	12
6.8	Table 9: Does financial advice improve diversification and portfolio performance?	12
6.9	Table 10: Does advice benefit the average advisee?	13
6.10	Table 11: Does the average advisee benefit if partially following?	14
6.11	Table 12: Who would most benefit from the advice?	15

7	Short summary	16
8	Conclusion	16
8.1	Core conclusion	16
8.2	Economic intuition	17
8.3	Incremental contribution revisited	17
8.4	Policy implications	17
8.5	Limitations	17
8.6	Takeaway	17

1 Big picture

- **Question.** Does free, unbiased, theoretically sound financial advice improve retail investors' portfolio choices once offered in the field?
- **Setting.** A large German brokerage randomly offered a new advice service to 8,195 active retail customers. The advice was free, algorithmic, and not tied to commissions or product sales.
- **Main demand-side result.** Only about 5% of the eligible customers obtained advice. Those who obtained it were more likely to be male, older, wealthier, more financially sophisticated, and longer-tenured brokerage clients.
- **Main follow-through result.** Even among advisees, advice was barely followed. The mean initial degree of following was only 15.6%, and the average over the post-advice measurement period was only 21.1%.
- **Quality of advice.** The recommendations were genuinely diversification-improving: they reduced single-stock exposure, reduced home bias, increased fund/ETF exposure, diversified across asset classes and regions, lowered HHI and idiosyncratic variance, and increased Sharpe ratios if fully implemented.
- **Impact on average advisee.** The average advisee did not substantially improve diversification and only modestly improved performance, mainly because most did not act on the advice.
- **Who needs advice?** The investors predicted to be least likely to obtain advice were also the ones for whom advice would have produced larger gains. This creates a targeting problem for policy.
- **Policy takeaway.** Removing supply-side conflicts is necessary but not sufficient. Advice must be both available and actually demanded and implemented by the investors who need it most.

2 Incremental contribution of the paper

- **Moves beyond conflicted-advice evidence.** Prior work often shows that advisors may be biased, conflicted, or poor performers. This paper studies a cleaner counterfactual: advice that is free, unbiased, and algorithmically designed to improve portfolio efficiency.
- **Links recommendations to post-advice behavior.** The paper observes the actual recommendation made to each client and the client's subsequent daily trading records, allowing a direct test of whether advice is obtained and followed.
- **Separates three channels.** It distinguishes whether advice is *good*, whether clients *take* it, and whether clients *follow* it. This decomposition is the core incremental insight.
- **Provides a demand-side regulatory critique.** Regulations that remove conflicts or improve disclosure solve the supply side of advice, but not the behavioral demand-side problem that low-sophistication investors may ignore or avoid advice.
- **Creates a field-study benchmark.** The study uses a real brokerage, real portfolios, real recommendations, and real follow-up trading behavior rather than hypothetical surveys or laboratory portfolio choices.
- **Measures implementation directly.** The degree-of-following measure converts portfolio overlap and recommended transactions into a quantitative measure of advice implementation.
- **Identifies a treatment-targeting mismatch.** The people most likely to benefit are least likely to demand advice, so policy cannot rely only on making unbiased advice available.

3 Data and measurement

3.1 Field setting and timeline

- The brokerage sent the advice offer to a random sample of 8,195 active retail customers in May 2009.
- The pre-advice period runs from September 2005 to May 2009, indexed as $t = -44$ to $t = 0$.
- Customers could obtain advice between May and October 2009. The first recommendation was given on May 13, 2009.
- The post-advice period runs until March 31, 2010, approximately $t = 11$ months.

Interpretation: The random offer creates a clean sample frame, but the key decision to obtain advice is endogenous. Much of the paper asks who opts in and what happens after opt-in.

3.2 Advice algorithm and recommended portfolios

- The optimizer considers the client’s existing portfolio, risk aversion, and available assets.
- It recommends diversified portfolios using securities for which the brokerage charges no commissions, limiting supply-side conflicts.
- It usually recommends a broad shift away from single stocks toward funds, ETFs, fixed income, commodities, and real estate.
- The recommended risky portfolio is large relative to the client account, but not generally infeasible relative to the client’s financial wealth.

Interpretation: The paper first establishes that the advice is normatively sound. Without that step, lack of follow-through could be rational skepticism about bad advice.

3.3 Degree of following

Let A_i denote the euro value of the actual portfolio, R_i the euro value of the recommended portfolio, and O_i the euro value of overlapping securities between the two. The paper’s degree of following is:

$$Follow_i = \frac{O_i}{A_i + R_i - O_i}.$$

- $Follow_i = 0$ if the actual and recommended portfolios do not overlap.
- $Follow_i = 1$ if the actual and recommended portfolios are identical.
- The paper also tracks the change in $Follow_i$ after the recommendation.

Interpretation: This measure is stricter than simply asking whether a client bought one recommended security. It measures portfolio-level implementation relative to all positions that would need to change.

3.4 Portfolio-efficiency outcomes

- **HHI:** Herfindahl-Hirschman index of portfolio weights, measuring concentration.
- **Idiosyncratic variance share:** the share of portfolio variance not explained by the Carhart four-factor model.
- **Sharpe ratio:** risk-adjusted performance.

- **MPPM:** manipulation-proof performance measure, a performance metric less sensitive to return manipulation.

Interpretation: The paper uses both diversification metrics and performance metrics. This matters because advice can improve diversification even if realized returns over the short post-period are noisy.

3.5 Investor characteristics

The paper uses demographics, account history, portfolio holdings, and behavior:

- gender, age, and microgeographic wealth status;
- account opening date and length of relationship;
- risky portfolio value, cash, risky share, tax-free asset share;
- trading intensity, turnover, disposition effect, home bias;
- portfolio performance measures before and after advice.

Interpretation: These covariates help explain both advice take-up and advice implementation. They also help test whether financially unsophisticated or poorly diversified investors are the ones who actually seek advice.

4 Models and empirical specifications

4.1 Probit model for obtaining advice

The participation equation is:

$$Pr(Advice_i = 1) = \Phi(\alpha + X_i'\beta),$$

where $Advice_i$ equals one once a client obtains the advisory service.

Interpretation: The coefficients identify which observable traits predict demand for advice. The key finding is that clients who are already more sophisticated, wealthier, and more connected to the brokerage are more likely to opt in.

4.2 OLS model for following advice

For clients who obtain advice, the paper estimates:

$$\Delta Follow_i = \alpha + X_i'\gamma + \varepsilon_i,$$

where $\Delta Follow_i$ is the increase in degree of following from the recommendation date to later horizons.

Interpretation: This regression asks why advisees do or do not implement the recommendation. The striking result is that few observable variables explain follow-through; many obvious candidates such as trust proxies, financial sophistication, or size of recommended change do not strongly predict implementation.

4.3 Advice-quality regression

To test whether the recommendation is sound, the paper compares recommended portfolios for the treatment group with actual portfolios for the control group:

$$\Delta Y_i = \alpha + \beta Advice_i + X_i' \delta + u_i,$$

where Y_i is HHI, idiosyncratic variance share, Sharpe ratio, or MPPM. For the treatment group, post-advice Y_i is computed using the recommended portfolio.

Interpretation: A positive coefficient for reduced concentration, reduced idiosyncratic variance, or increased Sharpe ratio means the advice would improve efficiency if implemented.

4.4 Average-advisee regression

The paper then repeats the same design but uses the actual post-advice portfolios of the treatment group:

$$\Delta Y_i^{actual} = \alpha + \beta Advice_i + X_i' \delta + u_i.$$

Interpretation: This identifies what actually happened to the average advisee. The contrast with the advice-quality regression isolates implementation failure.

4.5 Partial-following regression

The final step adds the increase in the degree of following:

$$\Delta Y_i^{actual} = \alpha + \beta Advice_i + \theta \Delta Follow_i + X_i' \delta + u_i.$$

Interpretation: If θ is significant, even partial implementation improves portfolio outcomes. The paper finds that partial following improves some efficiency measures, especially HHI and Sharpe ratio.

5 Identification and empirical design

5.1 What makes the field study useful

- The advice offer goes to a random sample of brokerage customers.
- The advice is free and not commission-driven, removing a major supply-side confound.
- The authors observe the actual recommendation and post-advice trading, which permits an implementation test.
- A control group of eligible customers who did not obtain advice helps evaluate counterfactual portfolio changes.

Interpretation: The design is strongest for documenting the behavioral sequence: offer $\rightarrow$ take-up $\rightarrow$ following $\rightarrow$ portfolio outcomes.

5.2 Main identification limits

- Obtaining advice is not randomized. The offer is random, but customers self-select into advice.
- Follow-through is also endogenous. More motivated or attentive clients may both follow advice and improve portfolios.
- The post-advice period is short, so realized returns are noisy.
- The sample is from one large German brokerage and may not fully generalize to all retail markets.

Interpretation: The strongest causal claim is not that advice mechanically causes better outcomes for all. The robust lesson is that unbiased advice can be sound yet fail as a policy remedy because take-up and implementation are weak.

5.3 Policy interpretation

Supply-side regulation can improve advice quality by reducing conflicts of interest, but the paper shows that this is not enough. The low take-up rate and low follow-through rate imply that demand-side frictions - attention, trust, inertia, complexity, overconfidence, and preferences for existing holdings - can dominate.

6 Tables and figures

6.1 Table 1: Advisees' original portfolios versus recommended portfolios

Read:

- The recommended portfolios cut single-stock exposure from 53% to 27% and raise funds from 26% to 57%.
- The equity share falls from 73% to 59%, while fixed income, commodities, and real estate all rise.
- German equity exposure falls from 52% to 30%, while Europe and Asia-Pacific shares rise.
- The recommended portfolio is roughly the same size as the original portfolio for the median client, but covers a large fraction of brokerage wealth and total wealth.

Economic message: The recommendation is not cosmetic. It asks clients to make large portfolio changes away from concentrated domestic stock holdings and toward diversified multi-asset portfolios.

Table 1
Advisees' original portfolios versus recommended portfolios

Instrument	Portfolio share	
	Original (%)	Recommended (%)
Single stocks	53	27
Funds	26	57
ETFs	4	10
Single bonds	4	0
Others	13	6
Total (%)	100	100

Asset class	Portfolio share	
	Original (%)	Recommended (%)
Equity	73	59
Fixed income	6	15
Commodities	1	13
Real estate	1	8
Others	19	5
Total (%)	100	100

Region	Equity share	
	Original (%)	Recommended (%)
Germany	52	30
Europe	14	24
North America	12	10
Asia Pacific	9	18
World	8	9
Central and eastern Europe	2	7
Other emerging markets	3	2
Total (%)	100	100

Ratio	Percentile						Standard deviation	N
	Mean	Median	5th	25th	75th	95th		
Recommended portfolio over original portfolio	1.19	1.00	0.92	0.99	1.04	1.88	0.91	380
Recommended portfolio over financial wealth with the brokerage	0.90	0.90	0.43	0.75	0.99	1.29	0.43	365
Recommended portfolio over total financial wealth	0.51	0.48	0.15	0.32	0.67	0.97	0.26	318
Recommended portfolio over total wealth	0.28	0.21	0.05	0.12	0.40	0.77	0.23	318

Panel A shows the portfolio share by instrument of the advisees' *original* portfolio compared with the portfolio *recommended* to the advisees at the time of the recommendation. Panel B shows the portfolio share by asset class of the advisees' *original* portfolio compared with the portfolio *recommended* to the advisees at the time of the recommendation. Panel C provides a regional breakdown of the equity share for the advisees' *original* portfolio compared with the portfolio *recommended* to the advisees at the time of the recommendation. Panel D compares

Table 1: Advisees' original portfolios versus recommended portfolios.

6.2 Table 2 and Figure 1: recommended asset classes by risk aversion, and field-study timeline

Read:

- Table 2 shows that recommended equity shares rise from 38% for the most risk-averse clients to 63% for the least risk-averse clients.
- Fixed income and money market shares move in the opposite direction, consistent with risk tailoring.
- Figure 1 shows the pre-advice period, offer window, first recommendation, and post-advice measurement period.
- The design observes long pre-advice histories and about eleven months of post-advice portfolio behavior.

Economic message: The optimizer is both diversified and individualized. The timeline also makes clear that the authors observe behavior before and after the recommendation.

Table 2
Average asset class shares of recommended portfolios by advisees' risk aversion

Asset class	Risk aversion			
	1 = Highest	2	3	4 = Lowest
Equity	38	43	53	63
Fixed income	27	20	14	7
Money market	14	15	13	10
Commodities	5	9	10	13
Real estate	14	10	7	5
Other	2	4	3	3
Total	100	100	100	100
N	20	95	172	55

Table 2 shows the average asset class share within advisees' recommended portfolios for each level of risk aversion.

(a) Table 2: Average asset class shares by risk aversion.

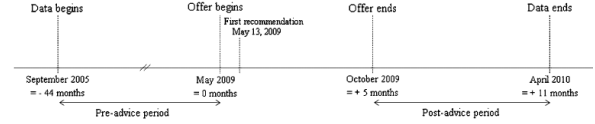


Figure 1
Timeline

The sequence of events in the field study (dates are always at the beginning of the respective month).

(b) Figure 1: Timeline.

6.3 Table 3 and Table 4: data collected and measurement of the degree of following

Read:

- Table 3 lists client demographics, portfolio characteristics, market data, recommended securities, and daily trading records.
- The data include both time-invariant demographics and monthly/daily portfolio measures.
- Table 4 illustrates the degree-of-following measure with an actual recommendation.
- The example has an overlap of EUR 4,224 and a degree of following of only 10%, even though some securities overlap.

Economic message: The paper's empirical strength is that it observes actual advice and actual implementation. Table 4 makes implementation failure concrete at the position level.

Type of data	Type of data	Dates available
Client demographics	Gender	Time-invariant
	Date of birth (measure of age)	Time-invariant
	Microgeographic status (measure of wealth)	Time-invariant
	Total financial and nonfinancial wealth (advisees only)	Time-invariant
	Risk aversion	Time-invariant
Portfolio characteristics	Actual position statements	Monthly
	Actual transactions	Daily
	Recommended securities	On day of recommendation
	Recommended transactions	On day of recommendation
	Cash	On day of recommendation
	Number of trades	Daily
Market data	Account opening date (measure of length of relationship)	Time invariant
	Carhart (1997) four factors on broad domestic index	
	–Market factor	Weekly
	–Small minus big (SMB)	Weekly
	–High minus low (HML)	Weekly
	–Momentum factor	Weekly
	Individual security returns	Daily

Table 3 summarizes the data collected during the course of the study. Client demographics and portfolio characteristics have been provided by the brokerage. The record date is July 2010. Market data are taken from the Thomson Reuters Datastream. The third column reports the availability of time-series data and their frequency.

(a) Table 3: Data collected.

Table 4 Illustration of the degree of following							
Original portfolio ($t = 0$)	€	Recommended portfolio ($t = 0$)	€	Required action	Buy €	Sell €	Kn €
Deutsche Bank	8,646			→		Sell	8,646
HSBC Indian Equity Fund	3,622			→		Sell	3,622
Raffisen CEE Equity Fund	2,792			→		Sell	2,792
HSBC IRIC Equity Fund	1,862			→		Sell	1,862
Commerzbank	439			→		Sell	439
E.ON	3,543	E.ON AG	681	→		Decrease	1,842
H&M	3,543	H&M	3,543	→	Buy		3,543
		Comstage ETF EONIA	4,072	→	Buy		4,072
		Schroder ISF Europa Corporate Bond Fund	3,883	→	Buy		3,883
		Allianz Pimco Europacrisis Bond Fund	3,799	→	Buy		3,799
		Allianz Pimco Corporate Bond Europa Fund	2,434	→	Buy		2,434
		UBS Lux Bond Fund	1,751	→	Buy		1,751
		Grontheve Europa Real Estate Fund	1,470	→	Buy		1,470
		Pictet EM Fund	1,247	→	Buy		1,247
		Allianz RCM Small Cap Fund	808	→	Buy		808
Total	23,426		23,688		19,463	19,202	4,224

Table 4 provides an actual example of the measure that we call the degree of following. Securities E.ON and H&M overlap between the actual and the recommended portfolio. The value of the overlap is 4,224 euros (€681 in E.ON and €3,543 in H&M). We calculate the degree of following as the overlap between the two portfolios divided by the value of assets in euros (i.e. the actual portfolio plus assets in the recommended portfolio, less the overlap (there: $4,224 / (23,426 + 23,688 - 4,224) = 10\%$)).

(b) Table 4: Illustration of the degree of following.

6.4 Table 5: Summary statistics

Read:

- Advice takers are 91.4% male versus 81.3% among non-takers.
- Advice takers are older, wealthier, have longer relationships with the bank, and hold larger risky portfolios.
- They trade somewhat more but have lower average turnover than non-takers.
- Recommended portfolios have much lower concentration and idiosyncratic variance than actual portfolios.

- Recommended portfolios also display higher Sharpe ratios and MPPM than actual portfolios during the post-advice window.

Economic message: The clients who obtain advice are not the least sophisticated investors. Moreover, the recommended portfolios look substantially more efficient than the actual portfolios.

Summary statistics								
Data variable	Measurement units	Obtain advice			Not obtain advice			t -test P -val
		Mean	Median	N	Mean	Median	N	
Client demographics								
Gender	Dummy = 1 if male	91.4	100.0	385	81.3	100.0	7,810	.00
Age	Years	52.9	52.0	385	49.0	47.0	7,810	.00
Wealth	Microgeographic status	6.6	7.0	340	6.3	6.0	6,847	.00
Portfolio characteristics								
Length of relationship with the bank	Years since account opening	9.1	9.7	385	7.4	8.8	7,810	.00
Risky portfolio value at $t = 0$	€ thousands	70.8	46.3	369	45.5	31.6	7,116	.00
Cash at $t = 0$	€ thousands	20.7	10.1	369	21.7	13.2	7,116	.52
Risky share at $t = 0$	%	75.9	85.4	369	66.1	73.9	7,116	.00
Average trades from $t = -44$ to $t = 0$	Trades per month	2.4	1.4	385	1.9	1.0	7,810	.01
Average portfolio turnover from $t = -44$ to $t = 0$	% monthly	5.8	3.6	378	7.5	4.0	7,745	.00
Disposition effect from $t = -44$ to $t = 0$	PQR - PQL	10.2	5.3	385	12.3	2.7	7,810	.13
Share of tax-free assets at $t = 0$	%	85.8	96.9	369	85.5	98.9	7,081	.82
Portfolio performance measures								
Raw returns								
From $t = -44$ to $t = 0$ (actual portfolios)	%, annualized	-5.3	-4.7	316	-7.0	-5.4	5,232	.13
From $t = 5$ to $t = 11$ (actual portfolios)	%, annualized	21.2	0.2	382	17.0	20.2	7,157	.21
From $t = 5$ to $t = 11$ (they and hold portfolios)	%, annualized	18.4	18.5	384				.08
From $t = 5$ to $t = 11$ (recommended portfolios)	%, annualized	24.8	22.6	383				.02
Standard deviation								
From $t = -44$ to $t = 0$ (actual portfolios)	%, annualized	25.8	23.8	316	30.4	26.5	5,232	.00
From $t = 5$ to $t = 11$ (actual portfolios)	%, annualized	15.0	14.0	382	21.2	17.9	7,157	.00
From $t = 5$ to $t = 11$ (they and hold portfolios)	%, annualized	14.5	14.0	384				.00
From $t = 5$ to $t = 11$ (recommended portfolios)	%, annualized	9.6	8.9	383				.00
Sharpe ratio								
From $t = -44$ to $t = 0$ (actual portfolios)	%	-4.1	-4.8	316	-4.0	-4.3	5,231	.69
From $t = 5$ to $t = 11$ (actual portfolios)	%	21.8	22.5	382	15.8	16.5	7,137	.00
From $t = 5$ to $t = 11$ (they and hold portfolios)	%	13.9	18.7	384				.02
From $t = 5$ to $t = 11$ (recommended portfolios)	%	35.2	35.4	384				.00

(a) Table 5: Summary statistics, part 1.

Data variable	Measurement units	Obtain advice			Not obtain advice			<i>t</i> -test <i>P</i> -val
		Mean	Median	<i>N</i>	Mean	Median	<i>N</i>	
MPPM								
From <i>t</i> = -44 to <i>t</i> = 0 (actual portfolios)	%	-16.0	-13.3	316	-22.3	-14.6	5,232	.06
From <i>t</i> = 5 to <i>t</i> = 11 (actual portfolios)	%	17.3	17.5	382	9.4	16.0	7,157	.03
From <i>t</i> = 5 to <i>t</i> = 11 (they and hold portfolios)	%	14.4	15.0	384				.18
From <i>t</i> = 5 to <i>t</i> = 11 (recommended portfolios)	%	22.4	21.2	384				.00
Four-factor alpha								
From <i>t</i> = -44 to <i>t</i> = 0 (actual portfolios)	%, annualized	-6.0	-5.6	316	-8.5	-6.3	5,231	.03
Four-factor beta								
From <i>t</i> = -44 to <i>t</i> = 0 (actual portfolios)		0.8	0.8	316	0.9	0.9	5,231	.21
From <i>t</i> = 5 to <i>t</i> = 11 (actual portfolios)		0.5	0.5	381	0.7	0.7	7,080	.00
From <i>t</i> = 5 to <i>t</i> = 11 (they and hold portfolios)		0.6	0.6	384				.00
From <i>t</i> = 5 to <i>t</i> = 11 (recommended portfolios)		0.4	0.4	384				.00
Portfolio diversification measures								
HHI								
At <i>t</i> = 0 (original portfolios)	%	12.0	7.6	369	20.3	10.8	7,100	.00
At <i>t</i> = 11 (actual portfolios)	%	10.4	4.8	377	10.8	9.8	7,014	.00
At <i>t</i> = 11 (they and hold portfolios)	%	9.7	4.7	384				.00
At <i>t</i> = 11 (recommended portfolios)	%	2.9	1.6	384				.00
Idiosyncratic variance share								
From <i>t</i> = -44 to <i>t</i> = 0 (actual portfolios)	%	36.3	30.6	316	39.5	33.9	5,231	.01
From <i>t</i> = 5 to <i>t</i> = 11 (actual portfolios)	%	29.6	21.9	381	32.5	23.7	7,080	.03
From <i>t</i> = 5 to <i>t</i> = 11 (they and hold portfolios)	%	30.0	19.6	384				.06
From <i>t</i> = 5 to <i>t</i> = 11 (recommended portfolios)	%	21.2	18.1	384				.00

(continued)

(continue)

(b) Table 5: Summary statistics, part 2.

6.5 Figure 2 and Table 6: Initial degree of following and post-advice implementation

Read:

- Figure 2 shows that the initial overlap between actual and recommended portfolios is very low.
- The distribution of the increase in following has a large mass near zero, meaning most investors do not materially move toward the recommendation.
- Table 6 reports mean following of 15.6% at the advice start, 25.4% thirty days later, but only 16.4% at the end of the post-advice period.
- The average over the post-advice measurement period is 21.1%, and only 157 clients count as followers over the whole period.

Economic message: The central mechanism of failure is not bad advice but weak implementation. Some clients act briefly, but many do not stick with the recommendation.

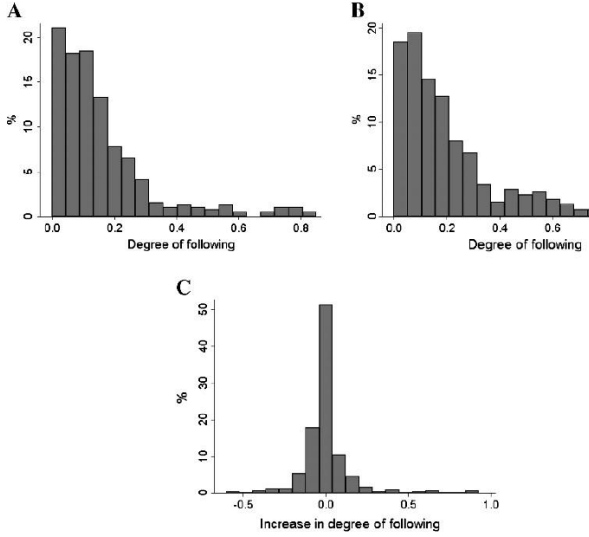


Figure 2
Initial degree of following
Panel A: The cross-sectional distribution of the degree of following when the first recommendation is received.
Panel B: The cross-sectional distribution of the average degree of following between time $t = \text{adv}$ and $t = 11$ months.
Panel C: Increase in degree of following from the time of the first recommendation to $t = 11$ months.

(a) Figure 2: Initial degree of following.

	$t = \text{Advice start}$	$t = 10 \text{ days}$	$t = 20 \text{ days}$	$t = 30 \text{ days}$	$t = 11 \text{ months}$ (end of measurement period)	Average over measurement period ($t = 5$ to $t = 11$ months)
Degree of following						
Mean (%)	15.6	21.6	24.4	25.4	16.4	21.1
Median (%)	10.8	14.2	15.5	16.2	10.4	14.7
Number of followers	n/a	98	141	158	125	157
Observations	385	385	385	385	385	385

Table 6 reports summary statistics for the degree of following. The table reports the mean degree of following, the median degree of following, the number of followers, and observations for six distinct time periods (upon receiving the first recommendation, after ten days, after twenty days, after thirty days, and at the end of the measurement period and the average over the entire measurement period). These time intervals are specific to each investor in the sample. Number of followers is defined as the number of investors who increase their degree of following until that point in time, at least marginally.

(b) Table 6: Summary statistics for the degree of following.

6.6 Table 7: Who opts for advice? A probit test

Read:

- Men and older investors are more likely to obtain advice.
- Larger risky portfolio value and longer brokerage relationships also predict take-up.
- More active trading predicts take-up, while turnover and disposition-effect measures often enter negatively.
- Low-wealth status reduces take-up; higher HHI and home bias also reduce take-up in key specifications.
- The pattern points to a selection problem: the investors most likely to need help are often less likely to request it.

Economic message: Advice demand is concentrated among investors who are already more engaged with the brokerage. The policy problem is that need and demand are not aligned.

Table 7
Who opts for advice? A probit test

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dummy male	0.361*** (0.000)	0.278*** (0.001)	0.283*** (0.002)	0.288*** (0.002)	0.291*** (0.000)	0.260*** (0.000)	0.233*** (0.007)
Age	0.000*** (0.000)	0.008*** (0.001)	0.011*** (0.000)	0.010*** (0.000)	0.008*** (0.000)	0.008*** (0.000)	0.011*** (0.000)
Dummy low wealth	-0.255* (0.059)	-0.267*** (0.047)	-0.177 (0.213)	-0.183 (0.197)	-0.264* (0.050)	-0.271** (0.041)	-0.200 (0.181)
Dummy high wealth	0.048 (0.353)	0.062 (0.235)	0.047 (0.413)	0.065 (0.417)	0.067 (0.217)	0.067 (0.201)	0.030 (0.615)
Log portfolio value ($t=0$)	0.144*** (0.000)	0.138*** (0.001)	0.136*** (0.004)	0.131*** (0.005)	0.102*** (0.008)	0.130*** (0.001)	0.138*** (0.002)
Length of relationship	0.062*** (0.000)	0.098*** (0.000)	0.098*** (0.000)	0.098*** (0.000)	0.064*** (0.000)	0.062*** (0.000)	0.042*** (0.000)
Trades per month ($t = -44$ to $t = 0$)	0.042*** (0.000)	0.043*** (0.000)	0.042*** (0.000)	0.038*** (0.000)	0.037*** (0.000)	0.037*** (0.000)	0.042*** (0.001)
Portfolio turnover ($t = -44$ to $t = 0$)	-0.116** (0.010)	-1.042* (0.080)	-1.030* (0.095)	-0.450 (0.110)	-0.966** (0.024)	-0.589 (0.437)	-0.287 (0.190)
Disposition effect ($t = -44$ to $t = 0$)	-0.161* (0.063)	-0.208** (0.011)	-0.203** (0.011)	-0.204** (0.027)	-0.154* (0.079)	-0.208** (0.009)	-0.287 (0.000)
Risky share ($t = 0$)	-0.103 (0.335)	-0.192 (0.115)	-0.201 (0.101)	-0.082 (0.468)	-0.103 (0.156)	-0.268** (0.029)	-0.287 (0.000)
Share of tax-free assets ($t = 0$)	-0.184* (0.085)	-0.219* (0.042)	-0.235* (0.044)	-0.207* (0.075)	-0.217* (0.055)	-0.223 (0.151)	-0.287 (0.000)
Alpha ($t = -44$ to $t = 0$)							
Idiosyncratic variance share ($t = -44$ to $t = 0$)							
HHI ($t = 0$)				-0.189 (0.241)			
					-0.883*** (0.000)		

(a) Table 7: Who opts for advice?, part 1.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Home bias ($t = 0$)						-0.197*** (0.008)	
Long-term raw return ($t = -44$ to $t = -1$)							27.917** (0.036)
Short-term raw return ($t = -1$ to $t = 0$)							-2.750* (0.080)
Constant	-3.937*** (0.000)	-3.932*** (0.000)	-4.327*** (0.000)	-4.366*** (0.000)	-3.513*** (0.000)	-3.850*** (0.000)	-4.287*** (0.000)
Observations	7,450	7,449	5,520	5,520	7,432	5,506	5,506
Pseudo- R^2	0.191	0.191	0.066	0.067	0.064	0.078	0.061

Table 7 reports probit estimates of the participation in the advisory model offered by the brokerage. Clients are set equal to one once they obtain the advice. For the estimation of the probit model, we include the following independent variables: a dummy that is equal to one if a client is male (*Dummy male*), the age of a client (*Age*), a dummy that is equal to one if a client falls into categories one to three of the microgeographic status rating by an external agency (*Dummy low wealth*), a dummy that is equal to one if a client falls into categories seven to nine of the microgeographic status (*Dummy high wealth*), the risky portfolio value of the customer (*Log portfolio value*), the number of years the client has been with the bank (*Length of relationship*), the number of trades per month (*Trades per month*), the average portfolio turnover per month (*Portfolio turnover*), the difference between the proportion of realized gains and losses (*Disposition effect*), the proportion of risky assets in the account (*Risky share*), the proportion of tax-free assets (*Share of tax-free assets*), the weekly alpha of a particular customer before opting for financial advice (*Alpha*), the idiosyncratic variance share (*Idiosyncratic variance share*), the Herfindahl-Hirschman index (*HHI*), the share of domestic equity (*Home bias*), the raw return from the beginning of observation until one month prior to the offer of financial advice (*Long-term raw return*), and the raw return of the month before the offer financial advice (*Short-term raw return*). Alpha and idiosyncratic risk share stem from the application of a Carhart (1997) four-factor model that is calibrated for Germany. P -values are in parentheses. R^2 values and number of observations are reported. R^2 values: (1) 0.191, (2) 0.191, (3) 0.066, (4) 0.067, (5) 0.064, (6) 0.078, (7) 0.061.

(b) Table 7: Who opts for advice?, part 2.

6.7 Table 8: Who chooses to follow advice?

Read:

- The dependent variable is the increase in degree of following after receiving advice.
- Low-wealth clients are less likely to follow; high-wealth clients are somewhat more likely to follow.
- Many expected predictors - sophistication, trust proxies, tax-free share, trading activity, and size of recommended change - do not explain much.
- The small explanatory power is consistent with the fact that most advice takers barely follow advice, leaving limited variation to explain.

Economic message: The paper can identify who demands advice better than who follows it. This is important because designing policies to increase follow-through is harder than removing advisor conflicts.

	Increase in degree of following					
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
Dummy male	-0.080 (0.122)	-0.076 (0.131)	-0.055 (0.280)	-0.062 (0.230)	-0.079 (0.119)	-0.080 (0.115)
Age	0.000 (0.535)	0.000 (0.536)	0.000 (0.995)	0.000 (0.967)	0.000 (0.872)	0.000 (0.925)
Dummy low wealth	-0.041** (0.046)	-0.054** (0.031)	-0.063*** (0.007)	-0.059** (0.018)	-0.053** (0.015)	-0.052** (0.009)
Dummy high wealth	0.045** (0.039)	0.047* (0.051)	0.038 (0.120)	0.038 (0.115)	0.047* (0.051)	0.043* (0.055)
Log portfolio value ($t = 0$)	-0.006*** (0.000)	-0.005*** (0.000)	-0.006*** (0.000)	-0.006*** (0.000)	-0.006*** (0.000)	-0.006*** (0.000)
Length of relationship	0.003 (0.390)	0.006 (0.356)	0.006 (0.188)	0.002 (0.459)	0.003 (0.425)	0.0 (0.475)
Trades per month ($t = -44$ to $t = 0$)	0.001 (0.751)	0.002 (0.656)	0.003 (0.490)	0.002 (0.614)	0.002 (0.638)	0.0 (0.658)
Portfolio turnover ($t = -44$ to $t = 0$)	-0.025 (0.069)	-0.025 (0.042)	-0.186 (0.470)	-0.054 (0.886)	-0.051 (0.811)	-0.1 (0.4)
Disposition effect ($t = -44$ to $t = 0$)	-0.027 (0.579)	-0.013 (0.798)	0.002 (0.969)	-0.026 (0.591)	-0.020 (0.538)	0.0 (0.2)
Risky share ($t = 0$)	0.002 (0.973)	-0.014 (0.753)	-0.018 (0.684)	-0.001 (0.976)	0.003 (0.945)	-0.0 (0.5)
Share of tax-free assets ($t = 0$)	0.051 (0.240)	0.054 (0.272)	0.064 (0.126)	0.057 (0.200)	0.057 (0.188)	0.0 (0.0)
Recommended portfolio over original portfolio	-0.003 (0.829)	-0.002 (0.841)	-0.003 (0.836)	-0.005 (0.728)	-0.003 (0.785)	0.0 (0.4)
Alpha ($t = -44$ to $t = 0$)			6.160 (0.155)			
Idiosyncratic variance share ($t = -44$ to $t = 0$)				0.089 (0.152)		
HHI ($t = 0$)					0.061 (0.534)	

(a) Table 8: Who chooses to follow advice?, part 1.

	Increase in degree of following					
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
Home bias ($t = 0$)						0.031 (0.382)
Long-term raw return ($t = -44$ to $t = -1$)						8.185 (0.169)
Short-term raw return ($t = -1$ to $t = 0$)						-0.872 (0.401)
Constant	0.754*** (0.000)	0.752*** (0.000)	0.676*** (0.000)	0.580*** (0.001)	0.741*** (0.000)	0.751*** (0.000)
Observations	369	365	313	313	365	297
R ²	0.132	0.139	0.138	0.139	0.141	0.131

Table 8 reports OLS estimates of the coefficients related to an increase in the degree of following measure from the date the investor received advice to $t = 20$ days (increase in *deg of following*). For the estimation of the model, we include the following independent variables: a dummy that is equal to one if a client is male (*Dummy male*), the age of a client (*Age*) a dummy that is equal to one if a client falls into categories one to three of a microgeographic status rating by an external agency (*Dummy low wealth*), a dummy that is equal to one if client falls into categories seven to nine of the microgeographic status (*Dummy high wealth*), the risky portfolio value of the customer (*Log portfolio value*), the number of years the client has been with the bank (*Length of relationship*), the number of trades per month (*Trades per month*), the average portfolio turnover per month (*Portfolio turnover*), the difference between the proportion of realized gains and losses (*Disposition effect*), the proportion of risky assets in the account (*Risky share*), the proportion of tax-free assets (*Share of tax-free assets*), a set of the required net investment (*Recommended portfolio/original portfolio*), the weekly alpha of a particular customer before opting for financial advice (*Alpha*), the idiosyncratic variance

(b) Table 8: Who chooses to follow advice?, part 2.

6.8 Table 9: Does financial advice improve diversification and portfolio performance?

Read:

- Table 9 evaluates what would happen if treatment clients implemented the recommended portfolios.
- The advice dummy is positive and significant for lower HHI and lower idiosyncratic variance.
- The advice dummy is also positive for performance metrics such as Sharpe ratio and MPPM.
- Thus, the recommendations are not merely different; they are measurably more efficient.

Economic message: Table 9 is the supply-side validation. The advice is sound. Any failure in realized outcomes must come from take-up or implementation, not from the quality of the recommendation.

Table 7
Does financial advice improve diversification and portfolio performance?

Dependent variable	Decrease in HHI				Decrease in idiosyncratic variance share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dummy advice	0.088*** (0.000)	0.091*** (0.000)	0.091*** (0.000)	0.091*** (0.000)	0.082*** (0.000)	0.087*** (0.000)	0.086*** (0.000)	0.086* (0.000)
Dummy male		-0.000 (0.993)	-0.000 (0.983)	-0.000 (0.959)		0.007 (0.309)	0.005 (0.523)	0.004 (0.549)
Age		-0.000 (1.000)	-0.000 (0.948)	-0.000 (0.871)		0.000 (0.687)	0.000 (0.765)	0.000 (0.796)
Dummy low wealth		0.003 (0.714)	0.004 (0.691)	0.004 (0.693)		-0.006 (0.618)	-0.008 (0.530)	-0.008 (0.525)
Dummy high wealth		0.003 (0.430)	0.003 (0.499)	0.002 (0.558)		0.008 (0.144)	0.009 (0.116)	0.009 (0.115)
Log portfolio value ($t = -44$)		-0.001 (0.698)	0.002 (0.266)	0.002 (0.292)		-0.018*** (0.000)	-0.020*** (0.000)	-0.019* (0.000)
Length of relationship			0.000 (0.673)	0.000 (0.753)			-0.001 (0.257)	-0.001 (0.257)
Trades per month ($t = -44$ to $t = 0$)			-0.000 (0.918)	-0.000 (0.938)			0.003** (0.042)	0.003* (0.038)
Portfolio turnover ($t = -44$ to $t = 0$)			-0.007 (0.924)	-0.002 (0.977)			0.113 (0.103)	0.098 (0.155)
Disposition effect ($t = -44$ to $t = 0$)			-0.009 (0.282)	-0.009 (0.287)			0.012 (0.228)	0.014 (0.165)
Risky share ($t = 0$)			-0.032*** (0.000)	-0.032*** (0.000)			0.038*** (0.000)	0.039* (0.000)
Share of tax-free assets ($t = 0$)			-0.020 (0.266)	-0.019 (0.301)			-0.046*** (0.009)	-0.045* (0.012)
Alpha ($t = -44$ to $t = 0$)				0.167 (0.901)				-1.549 (0.158)
Constant	0.002 (0.225)	0.006 (0.732)	0.018 (0.437)	0.020 (0.400)	0.072*** (0.000)	0.241*** (0.000)	0.274*** (0.000)	0.267* (0.000)
Observations	7,255	5,426	5,403	5,402	5,453	5,450	5,444	5,444
R^2	0.016	0.021	0.027	0.027	0.009	0.028	0.039	0.040

Table 9: Does financial advice improve diversification and portfolio performance?, part 1.

Table 9
Continued

Dependent variable	Increase in Sharpe ratio				Increase in MPPM			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Dummy advice	0.193*** (0.000)	0.189*** (0.000)	0.189*** (0.000)	0.189*** (0.000)	0.053*** (0.000)	0.057*** (0.000)	0.041*** (0.000)	0.083* (0.000)
Dummy male	-0.020*** (0.000)	-0.015*** (0.000)	-0.015*** (0.000)	-0.015*** (0.000)	0.015 (0.392)	0.002 (0.908)	-0.016 (0.317)	-0.016 (0.317)
Age	-0.001*** (0.000)	-0.000* (0.075)	-0.000* (0.075)	-0.000* (0.075)	-0.001 (0.345)	-0.001 (0.209)	-0.001* (0.050)	-0.001* (0.050)
Dummy low wealth	0.003 (0.181)	0.004 (0.246)	0.004 (0.246)	0.004 (0.246)	-0.006 (0.836)	-0.001 (0.962)	-0.008 (0.745)	-0.008 (0.745)
Dummy high wealth	0.003 (0.456)	0.004 (0.320)	0.004 (0.320)	0.004 (0.320)	0.023 (0.352)	0.024 (0.306)	0.028 (0.197)	0.028 (0.197)
Log portfolio value ($t = -44$)	0.010*** (0.000)	0.007* (0.062)	0.006** (0.021)	0.006** (0.021)	-0.008 (0.460)	-0.018 (0.161)	0.003 (0.839)	0.003 (0.839)
Length of relationship		-0.001 (0.161)	-0.001 (0.195)	-0.001 (0.195)		0.002 (0.816)	0.002 (0.816)	0.002 (0.816)
Trades per month ($t = -44$ to $t = 0$)		0.011*** (0.000)	0.011*** (0.000)	0.011*** (0.000)	0.017 (0.147)	0.026 (0.021)	0.026 (0.021)	0.026 (0.021)
Portfolio turnover ($t = -44$ to $t = 0$)		-0.513*** (0.000)	-0.513*** (0.000)	-0.513*** (0.000)	1.108** (0.029)	0.034 (0.960)	0.034 (0.960)	0.034 (0.960)
Disposition effect ($t = -44$ to $t = 0$)		0.008 (0.307)	0.012 (0.131)	0.012 (0.131)	-0.110* (0.001)	-0.110* (0.001)	-0.110* (0.001)	-0.110* (0.001)
Risky share ($t = 0$)		0.008 (0.252)	0.009 (0.169)	0.009 (0.169)	0.085 (0.215)	0.107 (0.107)	0.107 (0.107)	0.107 (0.107)

(continue)

Table 9
Continued

Dependent variable	Increase in Sharpe ratio				Increase in MPPM			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Share of tax-free assets ($t = 0$)			0.064*** (0.000)	0.065*** (0.000)		0.072 (0.396)	0.154* (0.038)	0.154* (0.038)
Alpha ($t = -44$ to $t = 0$)			-3.510*** (0.000)	-3.510*** (0.000)		-3.510*** (0.000)	-3.510*** (0.000)	-3.510*** (0.000)
Constant	0.199*** (0.000)	0.138*** (0.000)	0.146*** (0.000)	0.131*** (0.000)	0.330*** (0.000)	0.429*** (0.000)	0.365*** (0.003)	0.365*** (0.003)
Observations	5,479	5,476	5,470	5,469	5,487	5,484	5,478	5,477
R^2	0.079	0.091	0.141	0.148	0.000	0.001	0.019	0.024

(a) Table 9, part 2.

(b) Table 9, part 3.

6.9 Table 10: Does advice benefit the average advisee?

Read:

- Table 10 repeats the performance tests using the actual portfolios of treatment clients rather than the recommended portfolios.
- The average advisee does not materially improve diversification.
- Performance improvements exist in some specifications but are much smaller than the improvements implied by fully implementing the recommendation.
- The contrast with Table 9 isolates the implementation gap.

Economic message: Sound advice creates potential gains, but actual gains are modest because clients do not sufficiently alter their portfolios.

Table 10
Does the average advisee benefit?

Dependent variable	Decrease in HHI				Decrease in idiosyncratic variance share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dummy advice	0.013 (0.144)	0.011 (0.298)	0.010 (0.321)	0.011 (0.304)	-0.003 (0.807)	0.002 (0.886)	0.001 (0.907)	0.002 (0.858)
Dummy male		-0.001 (0.901)	-0.000 (0.961)	-0.000 (0.940)		0.005 (0.475)	0.003 (0.637)	0.003 (0.667)
Age		-0.000 (0.598)	-0.000 (0.619)	-0.000 (0.553)		0.000 (0.635)	0.000 (0.647)	0.000 (0.676)
Dummy low wealth		0.003 (0.713)	0.004 (0.689)	0.004 (0.691)		-0.005 (0.713)	-0.006 (0.608)	-0.006 (0.603)
Dummy high wealth		0.002 (0.681)	0.001 (0.743)	0.001 (0.814)		0.007 (0.189)	0.008 (0.153)	0.008 (0.151)
Log portfolio value ($t = -44$)		0.000 (0.899)	0.003 (0.131)	0.003 (0.151)		-0.017*** (0.000)	-0.019*** (0.000)	-0.019* (0.000)
Length of relationship			0.000 (0.756)	0.000 (0.839)			-0.002 (0.126)	-0.002 (0.126)
Trades per month ($t = -44$ to $t = 0$)			0.001 (0.390)	0.001 (0.376)			0.003** (0.017)	0.003* (0.015)
Portfolio turnover ($t = -44$ to $t = 0$)			-0.050 (0.464)	-0.045 (0.519)			0.077 (0.266)	0.062 (0.371)
Disposition effect ($t = -44$ to $t = 0$)			-0.009 (0.284)	-0.009 (0.283)			0.016 (0.132)	0.017* (0.089)
Risky share ($t = 0$)			-0.036*** (0.000)	-0.036*** (0.000)			0.035*** (0.000)	0.036* (0.000)
Share of tax-free assets ($t = 0$)			-0.017 (0.367)	-0.015 (0.407)			-0.037** (0.039)	-0.035* (0.048)
Alpha ($t = -44$ to $t = 0$)				0.243 (0.857)				-1.611 (0.144)
Constant	0.002 (0.225)	0.003 (0.853)	0.017 (0.479)	0.019 (0.432)	0.072*** (0.000)	0.232*** (0.000)	0.265*** (0.000)	0.258* (0.000)
Observations	7,251	5,422	5,399	5,398	5,453	5,450	5,444	5,444

Table 10: Does advice benefit the average advisee?, part 1.

Table 10
Continued

Dependent variable	Increase in Sharpe ratio				Increase in MPPM			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Dummy advice	0.060*** (0.000)	0.160*** (0.000)	0.056*** (0.000)	0.057*** (0.000)	0.010 (0.473)	0.014 (0.538)	-0.001 (0.927)	0.040* (0.027)
Dummy male		-0.021*** (0.000)	-0.015*** (0.000)	-0.016*** (0.000)	0.015 (0.354)	0.002 (0.999)	-0.016 (0.526)	-0.016 (0.560)
Age		-0.001*** (0.000)	-0.000* (0.056)	-0.000* (0.051)	-0.001 (0.231)	-0.001 (0.198)	-0.001 (0.046)	-0.001 (0.046)
Dummy low wealth		0.011 (0.288)	0.009 (0.377)	0.008 (0.395)	0.008 (0.365)	-0.007 (0.788)	-0.010 (0.914)	-0.010 (0.699)
Dummy high wealth		0.003 (0.513)	0.004 (0.364)	0.004 (0.317)	0.022 (0.365)	0.024 (0.318)	0.027 (0.318)	0.027 (0.307)
Log portfolio value ($t = -44$)		0.010*** (0.000)	0.004** (0.042)	0.005** (0.014)	-0.008 (0.449)	-0.018 (0.161)	-0.018 (0.161)	-0.018 (0.161)
Length of relationship			-0.001 (0.139)	-0.001 (0.169)	0.001 (0.143)	0.002 (0.143)	0.002 (0.143)	0.002 (0.143)
Trades per month ($t = -44$ to $t = 0$)			0.011*** (0.000)	0.011*** (0.000)	0.018 (0.021)	0.018 (0.021)	0.020* (0.021)	0.020* (0.021)
Portfolio turnover ($t = -44$ to $t = 0$)			-0.520*** (0.000)	-0.555*** (0.000)	1.116** (0.028)	1.116** (0.028)		0.043 (0.882)
Disposition effect ($t = -44$ to $t = 0$)			0.008 (0.322)	0.012 (0.144)	-0.233*** (0.000)	-0.233*** (0.000)		-0.112* (0.000)
Risky share ($t = 0$)			0.004 (0.537)	0.006 (0.401)	0.082 (0.234)	0.104 (0.119)		0.104 (0.119)

(continue)

Table 10
Continued

Dependent variable	Increase in Sharpe ratio				Increase in MPPM			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Share of tax-free assets ($t = 0$)			0.063*** (0.000)	0.065*** (0.000)			0.074 (0.381)	0.156* (0.035)
Alpha ($t = -44$ to $t = 0$)				-3.417*** (0.000)				-0.756* (0.000)
Constant	0.190*** (0.000)	0.138*** (0.000)	0.148*** (0.000)	0.135*** (0.000)	0.330*** (0.000)	0.433*** (0.000)	0.367*** (0.000)	-0.034 (0.813)
Observations	5,479	5,476	5,470	5,469	5,487	5,484	5,478	5,477

(a) Table 10, part 2.

(b) Table 10, part 3.

6.10 Table 11: Does the average advisee benefit if partially following?

Read:

- Table 11 adds the improvement in the degree of following as a right-hand-side variable.
- Partial following is associated with improvement in HHI and Sharpe ratio.
- The effects on idiosyncratic variance share and MPPM are weaker.
- The results imply that even incomplete implementation can help, but only if clients actually move toward the recommendation.

Economic message: The obstacle is not that recommendations are useless unless implemented perfectly. Partial implementation matters; the problem is that implementation is often minimal.

Does the average advisee benefit if the advice is partially followed?

Dependent variable	Decrease in HHI				Decrease in idiosyncratic variance share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Improvement of degree of following	0.136*** (0.002)	0.146*** (0.003)	0.147*** (0.003)	0.147*** (0.004)	0.113 (0.122)	0.103 (0.150)	0.119 (0.100)	0.118 (0.104)
Dummy advice	0.012 (0.177)	0.009 (0.392)	0.008 (0.422)	0.009 (0.403)	-0.004 (0.727)	0.001 (0.964)	0.000 (0.998)	0.001 (0.947)
Dummy male		-0.001 (0.906)	-0.000 (0.960)	-0.000 (0.939)		0.005 (0.479)	0.003 (0.645)	0.003 (0.675)
Age		-0.000 (0.625)	-0.000 (0.647)	-0.000 (0.580)		0.000 (0.619)	0.000 (0.628)	0.000 (0.657)
Dummy low wealth		0.003 (0.699)	0.004 (0.676)	0.004 (0.678)		-0.004 (0.720)	-0.006 (0.616)	-0.006 (0.610)
Dummy high wealth		0.002 (0.709)	0.001 (0.771)	0.001 (0.843)		0.007 (0.195)	0.008 (0.158)	0.008 (0.157)
Log portfolio value ($t = -44$)		0.000 (0.876)	0.003 (0.126)	0.003 (0.146)		-0.017*** (0.000)	-0.019*** (0.000)	-0.019* (0.000)
Length of relationship			0.000 (0.772)	0.000 (0.855)			-0.002 (0.121)	-0.002 (0.121)
Trades per month ($t = -44$ to $t = 0$)			0.001 (0.354)	0.001 (0.341)			0.003** (0.016)	0.003* (0.014)
Portfolio turnover ($t = -44$ to $t = 0$)			-0.050 (0.463)	-0.045 (0.519)			0.077 (0.266)	0.062 (0.370)
Disposition effect ($t = -44$ to $t = 0$)			-0.009 (0.272)	-0.009 (0.270)			0.015 (0.137)	0.017* (0.093)
Risky share ($t = 0$)			-0.036*** (0.000)	-0.036*** (0.000)			0.036*** (0.000)	0.036* (0.000)
Share of tax-free assets ($t = 0$)			-0.017 (0.342)	-0.016 (0.380)			-0.037** (0.036)	-0.036* (0.044)
Alpha ($t = -44$ to $t = 0$)				0.259 (0.848)				-1.603 (0.147)
Constant	0.002 (0.225)	0.002 (0.890)	0.017 (0.478)	0.019 (0.430)	0.072*** (0.000)	0.232*** (0.000)	0.265*** (0.000)	0.258* (0.000)
Observations	7,251	5,422	5,399	5,398	5,453	5,450	5,444	5,444
R^2	0.001	0.002	0.008	0.009	0.001	0.017	0.027	0.028

Table 11: Does the average advisee benefit if partially following?, part 1.

Table 11
Continued

Dependent variable	Increase in Sharpe ratio				Increase in MPPM			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Improvement of degree of following	0.192*** (0.001)	0.197*** (0.001)	0.187*** (0.001)	0.185*** (0.002)	0.018 (0.752)	0.006 (0.913)	0.085 (0.199)	0.026 (0.645)
Dummy advice	0.058*** (0.000)	0.054*** (0.000)	0.054*** (0.000)	0.055*** (0.000)	0.010 (0.622)	0.014 (0.477)	-0.002 (0.898)	0.007 (0.030)
Dummy male		-0.021*** (0.000)	-0.015*** (0.004)	-0.016*** (0.003)		-0.001 (0.384)	-0.001 (0.902)	-0.001* (0.125)
Age		-0.001*** (0.003)	-0.000* (0.063)	-0.000* (0.058)		-0.001 (0.232)	-0.001 (0.200)	-0.001* (0.047)
Dummy low wealth		0.011 (0.276)	0.009 (0.365)	0.009 (0.363)		-0.007 (0.789)	-0.003 (0.917)	-0.010 (0.700)
Dummy high wealth		0.003 (0.591)	0.004 (0.594)	0.004 (0.613)		0.022 (0.079)	0.024 (0.161)	0.027 (0.840)
Log portfolio value ($t = -44$)		0.010*** (0.000)	0.004** (0.009)	0.005** (0.013)		-0.008 (0.449)	-0.014 (0.001)	0.001 (0.902)
Length of relationship			0.011*** (0.128)	0.011*** (0.155)		0.018 (0.818)	0.020* (0.140)	0.020* (0.140)
Trades per month ($t = -44$ to $t = 0$)			0.011*** (0.000)	0.011*** (0.000)		0.018 (0.142)	0.020* (0.021)	0.020* (0.043)
Portfolio turnover ($t = -44$ to $t = 0$)			-0.520*** (0.000)	-0.555*** (0.000)		1.116** (0.028)	0.943 (0.082)	0.943 (0.082)
Disposition effect ($t = -44$ to $t = 0$)			0.008 (0.542)	0.012 (0.156)		-0.235*** (0.001)	-0.112* (0.000)	-0.104 (0.000)
Risky share ($t = 0$)			0.004 (0.522)	0.006 (0.389)		0.082 (0.244)	0.104 (0.118)	0.104 (0.813)
Share of tax-free assets ($t = 0$)			0.062*** (0.000)	0.064*** (0.000)		0.074 (0.384)	0.116* (0.006)	0.116* (0.006)

continue

Table 11
Continued

Dependent variable	Increase in Sharpe ratio				Increase in MPPM			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Alpha ($t = -44$ to $t = 0$)				-3.465*** (0.000)				-93.754* (0.000)
Constant	0.190*** (0.000)	0.137*** (0.000)	0.148*** (0.000)	0.133*** (0.000)	0.130*** (0.000)	0.413*** (0.000)	0.367*** (0.001)	-0.034 (0.813)
Observations	5,479	5,476	5,470	5,469	5,487	5,484	5,478	5,477
R^2	0.011	0.023	0.077	0.084	0.000	0.001	0.019	0.203

Table 11 reports OLS estimates of the coefficients related to a decrease in HHI (models 1 to 4) and a decrease in idiosyncratic variance share (models 5 to 8), an increase in Sharpe ratio (models 9 to 12) and an increase in MPPM (models 13 to 16). HHI and portfolio returns are calculated based on the actual portfolios of investors. The focus of the table is on the variables

(a) Table 11, part 2.

(b) Table 11, part 3.

6.11 Table 12: Who would most benefit from the advice?

Read:

- The authors split clients into those predicted to obtain advice and those predicted not to obtain advice.
- Advice creates larger diversification gains for clients predicted not to obtain advice.
- This confirms a mismatch between demand for advice and benefits from advice.
- The result strengthens the public policy argument: access alone does not target the investors most in need.

Economic message: The paper's most important policy conclusion is not just that advice is underused, but that underuse is concentrated among those who would benefit most.

Table 12
Who would benefit most from advice?

Dependent variable	Decrease in HHI			Decrease in idiosyncratic variance share		
	Predicted to obtain advice	Not predicted to obtain advice	<i>P</i> -value	Predicted to obtain advice	Not predicted to obtain advice	<i>P</i> -value
Dummy advice	0.042*** (0.000)	0.099*** (0.000)	.00	0.066** (0.013)	0.077*** (0.000)	.73
Constant	−0.007* (0.089)	0.003 (0.135)		0.112*** (0.000)	0.070*** (0.000)	
Observations	366	6,859		359	5,085	
<i>R</i> ²	0.056	0.017		0.018	0.007	
Dependent variable	Increase in Sharpe ratio			Increase in MPPM		
	Predicted to obtain advice	Not predicted to obtain advice	<i>P</i> -value	Predicted to obtain advice	Not predicted to obtain advice	<i>P</i> -value
Dummy advice	0.147*** (0.000)	0.194*** (0.000)	.00	−0.034 (0.579)	0.053*** (0.006)	.18
Constant	0.250*** (0.000)	0.196*** (0.000)		0.439*** (0.000)	0.324*** (0.000)	
Observations	359	5,110		359	5,118	
<i>R</i> ²	0.179	0.069		0.000	0.000	

Table 12 reports coefficient estimates of the variable “dummy advice” when the tests of Table 9 are conducted on subgroups. The first subgroup consists of clients who are predicted to obtain advice. These are 5% of the 7,432 clients with the highest probability of obtaining the advice predictions as given in model 5 of Table 7. Their size is 372. Of these, sixty-two took advice and so have the indicator variable “dummy advice” turned on to equal one, and 310 did not take advice and so have the indicator variable “dummy advice” turned on to equal zero. The second subgroup consists of clients who are not predicted to obtain advice. These are 95% of the 7,432 clients with the lowest probability of obtaining the advice predictions as given in model 5 of Table 7. Their size is 7,060. Of these, 307 took advice and so have the indicator variable “dummy advice” turned on to equal one, and 6,753 did not take advice and so have the indicator variable “dummy advice” turned on to equal zero. *R*² values and number of observations are reported. *** denotes significance at 1% or less, ** significance at 5% or less and * significance at 10% or less. Heteroscedasticity robust standard errors are used. The third column reports the *p*-value of a test of equality of the coefficients for *Dummy advice* using seemingly unrelated estimation. Different counts of observations are due to data availability of certain variables (see Table 5); results are robust to using the lowest common denominator

Table 12: Subgroup tests - clients predicted to obtain versus not obtain advice.

7 Short summary

- A German brokerage offered unbiased, free, algorithmic financial advice to 8,195 randomly selected active retail customers.
- Only about 5% of clients obtained advice.
- The advice was high quality: recommended portfolios were more diversified, less home-biased, less concentrated, and better on risk-adjusted performance metrics.
- Advice takers were not the investors most in need; they were more likely to be older, male, wealthier, more experienced, and longer-tenured.
- Most advice takers barely followed the advice.
- The average degree of following was low and mostly did not rise persistently after recommendations.
- The average advisee improved little because implementation was weak.
- Partial following improved some outcomes, especially concentration and Sharpe ratio.
- The investors predicted not to demand advice were the ones who would benefit more from it.
- The paper’s core message is that unbiased advice solves the supply-side problem but not the demand-side problem.

8 Conclusion

8.1 Core conclusion

The paper shows that unbiased and theoretically sound financial advice is not sufficient to improve retail investor outcomes. The advice generated by the brokerage’s algorithm would have improved portfolio efficiency if implemented, but few eligible clients requested it and even fewer implemented it meaningfully.

8.2 Economic intuition

The key intuition is a three-stage failure. First, advice must be available and unbiased. The study satisfies this condition. Second, investors must demand advice. Most do not. Third, investors who receive advice must implement it. Most do not. Therefore, policy that focuses only on the quality and neutrality of advice misses a central demand-side friction.

8.3 Incremental contribution revisited

The paper’s incremental contribution is to transform the advice question from “is advice biased?” to “does unbiased advice work in practice?” It documents that the limiting constraint is not only supply-side bias but also take-up and follow-through. By observing actual recommendations and subsequent trades, the paper reveals that advice can be good, free, and unbiased yet still fail to change portfolios enough to deliver benefits.

8.4 Policy implications

The results caution against relying solely on disclosure, fiduciary duties, or conflict-of-interest rules. Such policies may improve advice quality, but they do not ensure that low-sophistication investors seek or follow advice. More active interventions may be needed, such as default portfolio changes, guided implementation, automatic rebalancing, opt-out advice, simpler recommendations, reminders, or commitment devices.

8.5 Limitations

The paper studies one brokerage and one advice program in Germany. Advice take-up and implementation may differ in other institutional settings, especially where advice is bundled with automatic execution or retirement defaults. Also, following advice is not randomly assigned, so the estimated effect of partial following should be interpreted as suggestive rather than fully causal.

8.6 Takeaway

The lasting takeaway is that household finance frictions are not solved by simply making good advice available. Advice must overcome investor attention, trust, inertia, overconfidence, complexity, tax concerns, and attachment to existing holdings. The paper therefore shifts the policy debate from advice quality alone to advice adoption and implementation.